

Abraham Teklu

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SUMMARY

Aspiring AI/ML Researcher and Physics Ph.D. with experience developing data-driven models and algorithms for complex systems. Expert in Python and C++ with a proven record of building robust data pipelines and automated simulation workflows using high-performance computing (HPC). Skilled at translating ambiguous research problems into actionable solutions using statistics, optimization, and prototyping novel algorithms. Eager to leverage a background in large-scale computational modeling toward machine learning driven decision systems.

EDUCATION

Stony Brook University, 2018 to 2026
Doctor of Philosophy, Ph.D. — Physics

- GPA: 3.67
- Awards: Turner Fellowship, Lourie Fellowship

Oregon State University, 2014 to 2018
Bachelor of Science, B.S. — Physics

- GPA: 3.80
- Awards: Academic Achievement Award, Nicodemus Scholar Award, Ken Krane Scholar Award

SKILLS

Technical Skills: Python, C++, ROOT, Linux Bash, Git, Docker, Singularity, CUDA, Matlab, HTML

Team Skills: Communication in Large Collaborations, Presentation Experience

Research Skills: Problem Solving, Data Analysis, Statistics, Algorithmic Programming

PROFESSIONAL EXPERIENCE

Neutron Reconstruction for T2K — Stony Brook University December 2021 – Present
Graduate Research Assistant

- Tools: ROOT, C++, Python
- T2K is a large collaboration of scientists, postdocs, and graduate students from around the world.
- Working on a team to develop data analysis tools to select neutrons in neutrino events.
- My current work improves neutrino energy calculations and the oscillation analysis fit, the main goal of T2K.

Construction of the Super-FGD — Stony Brook University October 2022 – December 2022
Graduate Research Assistant

- Joined 20 people for the T2K Near Detector Upgrade and assemble a detector made of 2 million scintillator cubes.
- Assembly took place at JPARC, a high-intensity proton accelerator facility located in Tokai, Japan.
- Completed the project and delivered the detector a month before the deadline.

T2K Oscillation Analysis — Stony Brook University January 2021 – December 2021
Graduate Research Assistant

- Tools: ROOT, C++, Docker, Singularity, CUDA
- Architected a modular containerized simulation framework for custom physics generator integration and large-parameter MCMC fits on NVIDIA GPUs, enabling first-ever joint T2K (500+ members, 12 countries) and NOvA (250+ members, 8 countries) oscillation analysis.

Programming Bootcamp — LISEA February 2023 – May 2023
Teacher / Mentor

- Tools: Python, Pandas, Matplotlib, Scikit-learn, Jupyter
- Created a 6-week data science curriculum for a small group of disenfranchised minorities. See more here, lisea.org
- Lectured and have 1-on-1 sessions to improve programming skill and help with assignments.

Novel 3D Scintillator Detector Prototypes — Stony Brook University August 2019 – August 2023
Graduate Research Assistant

- Tools: ROOT, C++, Python
- Created and executed relevant physics studies that led to new measurements of the neutron cross-section and doubled the energy range of the existing neutron cross-section measurements on scintillator.
- Worked with 61 researchers from 19 universities on the prototype construction at CERN (Geneva, Switzerland) and Stony Brook University (Stony Brook, NY). Led a data-taking team in Los Alamos National Lab (Los Alamos, NM) and was a major contributor to the analysis.
- The work produced 3 co-authored publications and an APS April Meeting Talk.

CERN — Stony Brook University June 2018 – August 2018
Visiting Research Assistant

- Tools: ROOT, C++, Python
- Assembled and tested the Super-FGD Prototype. The world's first detector that can do event-by-event neutron kinematic reconstruction. Ran the first tests with a charged particle beam on a detector of this type.

General Atomics DIII-D Tokamak Fusion Reactor June 2017 – August 2017
SULI Intern

- Tools: C++, Python
- Worked with 3 staff scientists at DIII-D, to model plasma in the DIII-D Tokamak Fusion Reactor.
- Contributed to a co-authored publication, and presented a poster at the APS DPH Meeting. Also contributed to 4 other co-authored talks and posters.

CIERA REU — Northwestern University June 2016 – August 2016
CIERA REU Intern

- Tools: MATLAB, HTML
- Found an approximate numerical solution for an unstable region of the Holling-Tanner model computationally.
- Presented findings in a poster at the American Astronomical Society (AAS) and made a website, [CIERA REU](https://ciereu.org).

Fermilab — Oregon State University January 2016 – January 2018
Undergraduate Research Assistant

- Tools: ROOT, C++, Python
- Modeled antineutrino interactions for the MINERvA detector at Fermilab.
- Analysis was included in a co-authored publication by the MINERvA collaboration.